

# Abhigyan Mohanta

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## EDUCATION

### Siksha 'O' Anusandhan University (ITER)

Nov 2022 – May 2026

*Bachelor of Technology in Computer Science Engineering*

*CGPA: 9.61*

## EXPERIENCE

### Fullstack Developer

Sep 2024 – Oct 2025

*Arthur Digital*

*Remote, Munich, Germany*

- Added **Multi LLM** support to **Arthur Vibe**, Arthur's asynchronous collaboration and alignment platform.
- Engineered custom media streaming pipelines using **FFmpeg**, **Puppeteer**, **WebSockets**, and **PubSub** systems to enable real-time, bidirectional communication between Arthur rooms and **Microsoft Teams**.
- Developed a **Slack and Jira MCP client** that automated Jira issue management, sent contextual task updates to Slack, and enabled conversational workflows with a Slack bot that maintained thread-level memory.
- Integrated **LLMs** to build voice agent bots, improving real-time conversational experience in virtual rooms.
- Contributed to the development of a **DataSync service** for keeping virtual room data up-to-date across distributed systems.
- Implemented real-time WebSocket communication layers in frontend clients for dynamic room state updates.
- Researched and prototyped A/V streaming solutions using **Kasm workspaces** for live client demos.

### Full Stack Engineer Intern

Sep 2023 – Aug 2024

*Super Effective*

*Remote, Cayman Islands, UK*

- Developed the **Select Threader Chrome Extension**, replacing an expensive Twitter bot by bypassing **Twitter API V2 pricing** of \$5000 per month.
- Set up CI/CD pipelines** for the Threader, enabling **biweekly releases**, and integrated **Sentry for performance monitoring**.
- Created an Elixir service called Twitter BreadCrumb Service using **OAuth V2** to increase user engagement.
- Migrated the threads **PostgreSQL** database table and wrote multiple **unit tests** to support this post type. Integrated thread pages into the **Next.js Web Client** and resolved login event issues.

## PROJECTS

### AdmitRL | Go, Python, PyTorch, Kubernetes, Kueue, KWOK, kind

2025 – Present

- Built a **Kueue-native admission control system** for distributed AI workloads, combining cohort-aware gang protection with a **PPO-based policy** for elastic, flavor-aware, and topology-aware admission decisions in shared GPU clusters.
- Created a trace-driven training and evaluation stack from the **Alibaba Lingjun GPU cluster trace**, with a custom RL environment, masked actor-critic PPO training, and live multi-seed benchmarking on **KWOK + kind** against upstream **BestEffortFIFO** and **StrictFIFO** baselines.
- Reduced **head-gang blocking by 75.3%** and **gang completion time by 65.1%** versus upstream BestEffortFIFO on the cohort benchmark, while eliminating **100%** of small-job bypass during gang blocking in 5/5 seeds.
- Improved the elastic benchmark by **55.8% lower elastic wait time**, **39.0% lower elastic completion time**, and **26.4% higher throughput** versus BestEffortFIFO, with consistent gains across 5-seed runs.

## SKILLS

**Languages:** Go, JS/TS, Python, Java, Elixir, SQL

**DevOps/Cloud:** Docker, Kubernetes, GitHub Actions, Jenkins, CircleCI, ArgoCD, Prometheus, Grafana, Loki, Ansible, Vault, AWS

## ACHIEVEMENTS & VOLUNTEERING

**Twilio Quest:** Won Twilio Quest in Global Hack Week, Sep 2022

**Open Source:** Contributed to various Open Source projects, including projects under CNCF and Uptime Kuma.

**Certifications:** **KCNA** (Kubernetes and Cloud Native Associate), Dec 03, 2025 among various others **Credly**